DO NOT WRITE ANYTHING IN THIS PORTION

Sant Dnyaneshwar Shikshan Sanstha's

Annasaheb Dange College of Engineering and Technology, Ashta

(An Empowered Autonomous Institute affiliated to Shivaji University, Kolhapur)

Sr. No.	224607
Examination	MSE ESE/MAKE UP
Program	UG / ~~PG~~
Dept./Specialization	CSE[IoT]
Class	FY / SY / TY / BTech
Academic Year	2026- 2027 Sem: Odd / ~~Even~~
Course Code	21CPC201
Course Name	DMTOC
Day & Date	9-10-26
Block No.	4
QP Code	QX-211

URN / Seat Number of Student									
1	0	2	6	1	0	2	0	0	1

Signature of the Student

BL

1026102001

t 104302 x 3266

Signature of the Invigilator

INSTRUCTIONS TO CANDIDATES TAKING THE EXAMINATION

Important Instruction

1. Only one answer book will be issued. **No supplement will be given.**
2. Write your **Seat Number / URN**, subject, course code, and semester clearly in the space provided on the title page.
3. Mention **question numbers clearly in the left margin only**.
4. Write on **both sides** of the pages. Rough work should be done in the space provided at the end of the answer book.
5. A warning bell will be given 10 minutes before the commencement of the examination. Candidates should take possession of their seats at that time. Another bell will be given at the beginning of the examination, and a final bell 10 minutes before the end. All writing must stop immediately after the final bell.
6. Use **black ink pen only**. Answers written in any other color or in pencil will not be evaluated (except for diagrams).
7. Do **not write your name, roll number, or any identification mark** anywhere inside the answer book.
8. The **candidate shall obtain the Junior Supervisor's signature** on the answer book at the specified space, both at the time of issue and submission of the answer book.
9. The answer book must be **handed over personally** to the Junior Supervisor before leaving the examination hall.
10. Candidates **shall not be permitted to leave** the examination hall during the period of examination.
11. **No books, notes, loose papers, log tables, mobile phones, smart watches, or electronic devices** are allowed in the examination hall.
12. **Maintain complete silence** in the examination hall. Any form of communication, verbal or non-verbal, will be treated as malpractice.
13. Candidates found **copying, possessing unauthorized material, or assisting others** will be immediately disqualified.
14. **Strike out** any rough work or cancelled answers neatly. Do not use whitener or erasers on written text.
15. **Follow all instructions** given by the Junior or Senior Supervisor during the examination.

विद्यार्थ्यांसाठी महत्त्वाच्या सूचना

1. केवळ एकच उत्तरपत्रिका दिली जाईल. पूरक उत्तरपत्रिका (Supplement) दिली जाणार नाही.
2. फक्त **काळ्या** शाईचा पेन **(Black Ink Pen Only)** वापरावा. इतर कोणत्याही रंगाची शाई किंवा पेन्सिल वापरू नये.
3. परीक्षेच्या सुरूवातीपूर्वी १० मिनिटांपूर्वी इशारा बेल दिली जाईल व शेवटच्या बेलनंतर लिखान ताबडतोब थांबवावे.
4. उत्तरपत्रिकेमध्ये नाव, रोल नंबर किंवा ओळख पटविणारी कोणतीही खूण लिहू नये.
5. ज्युनिअर सुपर्व्हायझरची स्वाक्षरी उत्तरपत्रिकेवर घेणे आवश्यक आहे.
6. परीक्षा चालू असताना परीक्षागृहाबाहेर जाता येणार नाही

Examiner (01)						
Que.	a)	b)	c)	d)	e)	Marks
01						
02						
03						
04						
05						
06						
Signature				Grand Total		

Examiner (02)						
Que.	a)	b)	c)	d)	e)	Marks
01						
02						
03						
04						
05						
06						
Signature				Grand Total		

Scrutiny / QC is done and found OK	

Scrutiny / QC is done and found OK	

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Q.No

Q.1]

a]

→ $(Q \wedge (P \rightarrow Q))$

P	Q	$P \rightarrow Q$	$Q \wedge (P \rightarrow Q)$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	F

$((P \rightarrow Q) \wedge (Q \rightarrow R)) \rightarrow (P \rightarrow R)$

P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	$P \rightarrow R$	$(P \rightarrow Q) \wedge (Q \rightarrow R)$	$((P \rightarrow Q) \wedge (Q \rightarrow R)) \rightarrow (P \rightarrow R)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	T	F	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	T	F	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

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Q.No.

$((P \to Q) \wedge (Q \to R)) \to (P \to R)$
T
T
T
T
T
T
T
T

∴ From above truth table $((P \to Q) \wedge (Q \to R)) \to (P \to R)$ is a tautology.

c]

→

i] → $P \to (Q \to R) \iff P \to (\sim Q \vee R)$

Let take $P \to (Q \to R)$

$\equiv \sim P \vee (Q \to R)$ .... $P \to Q = \sim P \vee Q$

$\sim P \vee (\sim Q \vee R)$ ... $P \to Q = \sim P \vee Q$

∴ $\sim P \vee \sim Q \vee R$ ..... ①

Q.No.

take

$P \rightarrow (\sim Q \vee R)$

$\sim P \vee (\sim Q \vee R)$ .... $P \rightarrow Q = \sim P \vee Q$

$\therefore \sim P \vee \sim Q \vee R$ ---- (2)

$\therefore$ from (1) & (2)

$P \rightarrow (Q \rightarrow R) \Leftrightarrow P \rightarrow (\sim Q \vee R)$

Hence proved.

2] $\rightarrow (P \rightarrow Q) \wedge (R \rightarrow Q) \Leftrightarrow (P \vee R) \rightarrow Q$

take

$(P \rightarrow Q) \wedge (R \rightarrow Q)$

$(\sim P \vee Q) \wedge (R \rightarrow Q)$ ... $P \rightarrow Q = \sim P \vee Q$

$(\sim P \vee Q) \wedge (\sim R \vee Q)$ ... $P \rightarrow Q = \sim P \vee Q$

$\therefore (Q \vee \sim P) \wedge (Q \vee \sim R)$ ... commutative law.

$\therefore Q \vee (\sim P \wedge \sim R)$ - (1) distribution law

take.

$(P \vee R) \rightarrow Q$

$\sim (P \vee R) \vee Q$ ... $P \rightarrow Q = \sim P \vee Q$

$(\sim P \wedge \sim R) \vee Q$ ... negation applied.

$\therefore Q \vee (\sim P \wedge \sim R)$ ... (2) commutative law.

$\therefore$ from (1) & (2) $(P \rightarrow Q) \wedge (R \rightarrow Q) \Leftrightarrow (P \vee R) \rightarrow Q$

Hence Proved.

Q.No.

Q2.

a]

→ Given: $P = \{5, 6, 7, 8\}$

$Q = \{6, 7\}$

$R = \{6, 7, 8, 9\}$

To solve:

$P - Q$

$P + Q$

$Q + R$

$(P+Q) + (Q+R)$

soln :-

$(P - Q) = \{5, 8\}$

$P + Q = (P - Q) \cup (Q - P)$

$\therefore Q - P = \{\}$ i.e ϕ

$\therefore P + Q = \{5, 8\} \cup \phi$

$\therefore P + Q = \{5, 8\}$

$\therefore (Q + R) = (Q - R) \cup (R - Q)$

$\therefore Q - R = \phi$

$R - Q = \{8, 9\}$

$\therefore (Q + R) = \phi \cup \{8, 9\}$

$\therefore (Q + R) = \{8, 9\}$

Q.No.

$(P+Q)+(Q+R) = ((P+Q)-(Q+R)) \cup ((Q+R)-(P+Q))$

$\therefore (P+Q)-(Q+R) = \{5\}$

$\therefore (Q+R)-(P+Q) = \{9\}$

$\therefore (P+Q)+(Q+R) = \{5\} \cup \{9\}$

$= \underline{\{5,9\}}$

$\therefore (P-Q) = \{5,8\}$

$\therefore (P+Q) = \{5,8\}$

$\therefore (Q+R) = \{8,9\}$

$\therefore (P+Q)+(Q+R) = \{5,9\}$

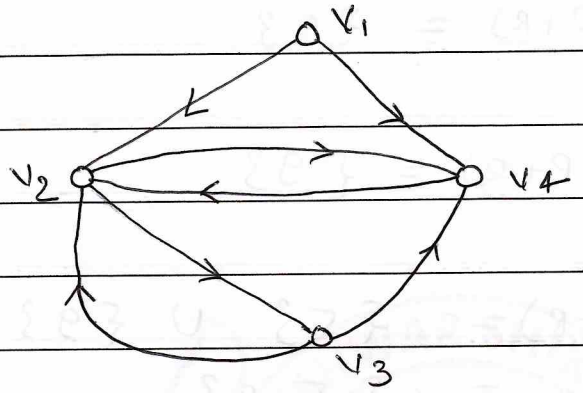
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Q.No.

Q3.

b]

→



adjacency matrix (A) =

	V_1	V_2	V_3	V_4
V_1	0	1	0	1
V_2	0	0	1	1
V_3	0	1	0	1
V_4	0	1	0	0

Path matrix (P) =

	V_1	V_2	V_3	V_4
V_1	0	1	1	1
V_2	0	1	1	1
V_3	0	1	1	1
V_4	0	1	1	1

Q.No.

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$A.A^T = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} (0+1+0+1) & (0+0+0+1) & (0+1+0+1) & (0+1+0+0) \\ (0+0+0+1) & (0+0+1+1) & (0+0+0+1) & (0+0+0+0) \\ (0+1+0+1) & (0+0+0+1) & (0+1+0+1) & (0+1+0+0) \\ (0+1+0+0) & (0+0+0+0) & (0+1+0+0) & (0+1+0+0) \end{bmatrix}$$

$$A.A^T = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 0 \\ 2 & 1 & 2 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Q.No.

$$A^T \cdot A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} (0+0+0+0) & (0+0+0+0) & (0+0+0+0) & (0+0+0+0) \\ (0+0+0+0) & (1+0+1+1) & (0+0+0+0) & (1+0+1+0) \\ (0+0+0+0) & (0+0+0+0) & (0+1+0+0) & (0+1+0+0) \\ (0+0+0+0) & (1+0+1+0) & (0+1+0+0) & (1+1+1+0) \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 2 & 1 & 3 \end{bmatrix}$$

$$A^2 = A \cdot A$$

$$= \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Q.No.

$$= \begin{bmatrix} (0+0+0+0) & (0+0+0+1) & (0+1+0+0) & (0+1+0+0) \\ (0+0+0+0) & (0+0+1+1) & (0+0+0+0) & (0+0+1+0) \\ (0+0+0+0) & (0+0+0+1) & (0+1+0+0) & (0+1+0+0) \\ (0+0+0+0) & (0+0+0+0) & (0+1+0+0) & (0+1+0+0) \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$A^3 = A^2 \cdot A$$

$$= \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} (0+0+0+0) & (0+0+1+1) & (0+1+0+0) & (0+1+1+0) \\ (0+0+0+0) & (0+0+0+1) & (0+2+0+0) & (0+2+0+0) \\ (0+0+0+0) & (0+0+1+1) & (0+1+0+0) & (0+1+1+0) \\ (0+0+0+0) & (0+0+1+1) & (0+0+0+0) & (0+0+1+0) \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 0 & 2 & 1 & 2 \\ 0 & 1 & 2 & 2 \\ 0 & 2 & 1 & 2 \\ 0 & 2 & 0 & 1 \end{bmatrix}$$

Q.No.

$$A^4 = A^3 \cdot A$$

$$= \begin{bmatrix} 0 & 2 & 1 & 2 \\ 0 & 1 & 2 & 2 \\ 0 & 2 & 1 & 2 \\ 0 & 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

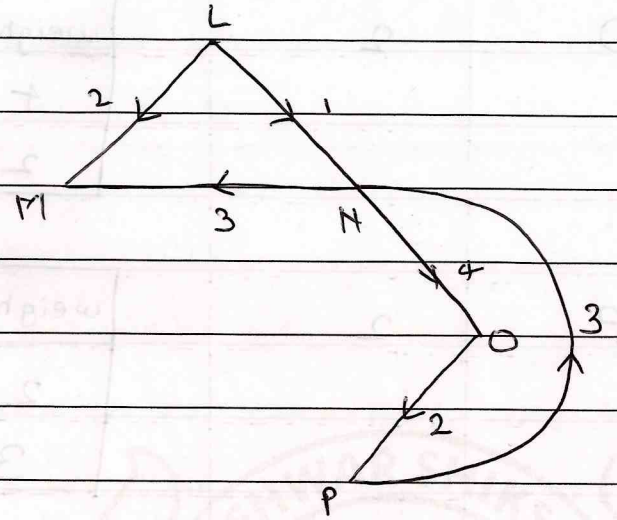
$$= \begin{bmatrix} (0+0+0+0)(0+0+1+2)(0+2+0+0)(0+2+1+0) \\ (0+0+0+0)(0+0+2+2)(0+1+0+0)(0+1+2+0) \\ (0+0+0+0)(0+0+1+2)(0+2+0+0)(0+2+1+0) \\ (0+0+0+0)(0+0+0+1)(0+2+0+0)(0+2+0+0) \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 0 & 3 & 2 & 3 \\ 0 & 4 & 1 & 3 \\ 0 & 3 & 2 & 3 \\ 0 & 1 & 2 & 2 \end{bmatrix}$$

Q.No.

c]

→



No of Node	node data	degree of node
1	L	2
2	M	2
3	N	3

Storage representation.

weight	destination
2	2
1	3

weight	destination
2	2
3	2

weight	destination
3	2
3	3
4	4

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Q.No.

4	0	2	
5	P	2	

weight	destination
4	4
2	5

weight	destination
2	5
3	3

Q2.

b]

→ $R = \{ (x, 2x) \mid x \in I \}$

$S = \{ (x, 3x) \mid x \in I \}$

Here $I = \{0, 1, 2, 3, \ldots\}$

$R \circ S = \{ \} \in \phi$ Here in this relation there is no transitive element so. $R \circ S = \phi$

Q.No.

$R \circ R = \phi$ ... Here in this relation there is no transitive element so $R \circ R = \phi$

$R \circ (R \circ R) = \phi$ ... Here in this relation $R \circ R$ is null so no transitive element

$\therefore R \circ (R \circ R) = \phi$

Q2.

b]

→ Given: $R = \{(x, 2x) \mid x \in I\}$

$S = \{(x, 3x) \mid x \in I\}$.

Let $I = \{1, 2, 3, 4, 5, 6, 7, 8\}$.

$\therefore R = \{(1,2)(2,4)(3,6)(4,8)\}$

$\therefore S = \{(1,3)(2,6)\}$

$\therefore R \circ S = \{(1,6)\}$

$\therefore R \circ R = \{(1,4)(2,8)\}$

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Q.No.

$R_o(R_oR) = \{ (2,8) (1,8) \}$

$(R_oS)_oR = \{ \phi$ Here is no transitive element so $(R_oS)_oR = \phi$

$\therefore R_oS = \{$

final ans :-

① $R_oS = \{(1,6)\}$

let $1 = x$

$\therefore R_oS = \{(x, 6x) \mid x \in I\}$

② $R_oR = \{(1,4) (2,8)\}$

consider $1 = x$ & $2 = x$

$R_oR = \{(x, 4x) \mid x \in I\}$

③ $R_o(R_oR) = \{(1,8)\}$

considering $1 = x$

$\therefore R_oR = \{(x, 8x) \mid x \in I\}$

④ $(R_oS)_oR = \phi$

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Q.No.

Q3.

a]

→ Given: $(\mathbb{Z}, +)$ is a group.

To Prove: $(2\mathbb{Z}, +)$ is a subgroup of $(\mathbb{Z}, +)$

To Prove ~~that~~ above ist we will have to prove $(2\mathbb{Z}, +)$ is a group for that $(2\mathbb{Z}, +)$ satisfy below condition: ① Closure
② Associative.
③ Identity.
④ Inverse.

① Closure

$\therefore (2\mathbb{Z}, +)$

~~let~~ Here $2\mathbb{Z}$ is set of even integer number.

$\therefore a+b \in 2\mathbb{Z}$

Here addition of any even no is even.

$\therefore$ Hence. $(2\mathbb{Z}, +)$ satisfy closure property.

② Associative.

$(a+b)+c = a+(b+c)$ ... associative property.

let $a = 2 \quad b = 4 \quad c = 6$

Q.No.

$(2+4)+6 = 2+(4+6)$

$6+6 = 2+10$

$12 = 12$

LHS = RHS

Hence $(\mathbb{Z}, +)$ satisfy associative property.

③ Identity.

$a+e=a$ ... Identity property.

let $a=4$

$4+e=4$

$e=4-4$

$e=0$.

∴ Hence $(\mathbb{Z}, +)$ satisfy Identity property.

④ Inverse

$a+a^{-1} =$

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